

Lab Evaluation:

Course Name: Object Oriented Programming				Course Code: CMPE 221L		
Assignment Type: Lab				Dated: 24 th October 2025		
Semester: 2 nd				Session: CYS 2025 Spring		
Lab/Project/Assignment #: 4				CLOs to be covered: CLO1		
Lab Title: Classes & Objects				Teacher Name: Engr. Afeef Obaid		
CLO1	Implement abstraction and encapsulation to develop reusable classes for objects of real-world problems					
Levels (Marks)	Level1	Level2	Level3	Level4	Level5	Level6
(10)						
Total						/10

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	9-10	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	7-8	L2	Submitted the lab tasks but have good understanding
Good	5-6	L3	Submitted the lab tasks but have weak understanding.
Basic	3-4	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1-2	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No. 4

Lab Goals/Objectives:

- Basic OO programming in java
- Objects and Classes

Equipment Required: Intel J IDEA Edition Community, Net Beans with JDK 13 Version, Visual Studio.

Theory/Relevant Material: Objects and classes, constructors, getter & setters, modifiers

1. Thinking of **Time** as an entity JTime class was created. Each object of this class represents a time-stamp (a mark in history). It has three attributes (hour, minute, second) and three constructors along with some other functions. Enhance its functionality by added following functions to it and test the functionality by updating the driver function.
 - a. Add a function that finds the time elapsed in seconds between two time-stamps.
 - b. Add a function that finds the time elapsed in two time-stamps in format of another time-stamp.
 - c. Add another constructor that takes total number of seconds elapsed after 12 am and construct the timestamp accordingly.
2. We are going to develop a menu-driven application to manage a simple bank account class. It supports the following operations
 - a. Deposit money (to add an amount in current balance)
 - b. Withdraw money (to subtract an amount from the current balance)
 - c. Check balance (to display current balance of customer)

Note: Use a Switch statement to enter the choice of operation from the user. For initially create a function withname saving account to store the current balance of the customer. Write a Java class along with a driver program to test it.

Something is Brewing:

3. So lately Ali has been walking as per your instructions. Avoiding obstacles in its path. Now Hasan wants to startwalking with Ali. But Hasan will walk randomly and will avoid the same obstacles as Ali. Write a code that will simulate both the walk patterns of Ali and Hasan simultaneously. Output can e of the form:

Starting: Ali's Location is = (0 : 0) Hassan's Location is = (0 : 0) Hasan has Random Movement, Ali follows order

ALi: Where should i go now: v Ali is at = (0 : -1)
Hasan is at = (0 : 1)

ALi: Where should i go now: ^ Ali is at = (0 : 0)
Hasan is at = (0 : 2)

ALi: Where should i go now: ^ Ali is at = (0 : 1)
Hasan encounters hurdle at (1:2) Hasan is at = (0 : 2)